ENTERPRISE OPERATIONS MANUAL • GROUNDED REFERENCE

AgriLeave System & User Guide

Complete operational manual, policy entitlement rules, multi-tier approvals, and Gemini AI assistant documentation for Agri Experience Limited.

ORGANIZATION
Agri Experience Limited

SYSTEM VERSION
AgriLeave v3.4.0 (Production)

BACKEND ARCHITECTURE
Google Apps Script & Drive DB

MASTER MIRROR
Google Sheets Dual-Store

Grounded System Documentation Notice

This manual is strictly compiled from active system code (`Code.js` , `Index.html` , and system settings). It documents exact entitlement formulas, multi-tier approval thresholds, deduction priorities, and AI parameters.

© 2026 Agri Experience Limited. All rights reserved. Confidential and proprietary software.

1. System Overview & Architecture

AgriLeave is an enterprise leave operations platform engineered specifically for **Agri Experience Limited**. Hosted natively on **Google Apps Script (GAS)**, the application operates entirely within the organization's Google Workspace ecosystem with zero external server infrastructure.

Dual-Store Database Architecture

Ultra-fast Google Drive JSON storage for sub-second web operations paired with an automatic, live Google Sheets Master Mirror for reporting and business intelligence.

Automated Accruals & Easter Expiry

Automated monthly accrual of 1.75 days (21 days/year) and dynamic Easter Sunday calculation for previous-year carryover expiration.

Multi-Tier Approval Routing

Single-tier supervisor approval for leaves ≤ 5 days; automatic secondary HR endorsement for extended (> 5 days) and specialized leaves.

Grounded Gemini AI Assistant

A floating, role-grounded conversational agent utilizing Google AI Studio's Gemini Flash model chain for policy answers and balance summaries.

Dual-Store Persistence Layer

Store Type	Physical Location / File	Data Entities & Operations
Primary Store	Google Drive JSON Folder: `Agri Experience Leave System/`	`staff_data.json` (profiles & balances), `leave_applications.json`, `compensation_claims.json`, `blackout_periods.json`, `gazetted_holidays.json`, `annual_reset_logs.json`, `rag_knowledge_base.json`.
Master Mirror	Google Sheets Workbook: `AgriLeave Master Database & Mirror`	Four live-synchronized worksheets: `Applications`, `Staff_Balances`, `Gazetted_Holidays`, and `Audit_Log`. Synchronous dual-write on every update.
Workspace APIs	Gmail API & Calendar API	Delivers HTML notification emails with 1-click approvals, configures Gmail Vacation Responder (Out-of-Office), and writes events to Google Calendar.
AI Intelligence	Google AI Studio (Gemini Chain)	Free-tier fallback chain: `gemini-3.5-flash-lite` → `gemini-3.5-flash` → `gemini-3.7-flash` → `gemini-3.8-flash` with `text-embedding-004`.

2. Interface Overview: Main Dashboard

Upon logging in via Google Workspace Single Sign-On (SSO), employees are greeted with their personalized **Dashboard**. The dashboard provides an instantaneous overview of available leave balances, upcoming public holidays, quick actions, and recent activity.

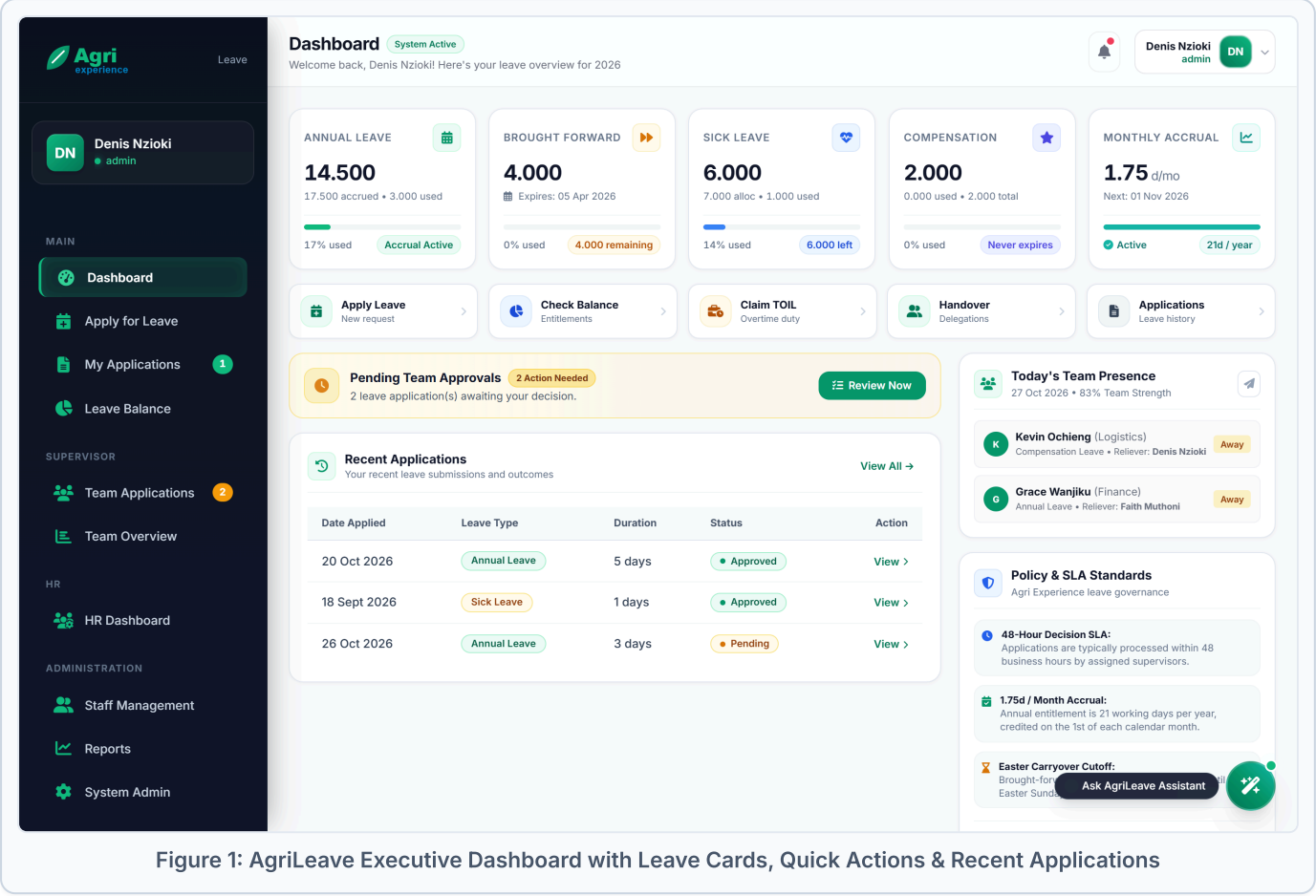


Figure 1: AgriLeave Executive Dashboard with Leave Cards, Quick Actions & Recent Applications

Key Dashboard Elements (Grounded in AppState)

- **Annual Leave Balance Card:** Real-time display of available working days, cumulative year-to-date accruals (`accruedYearToDate`), and consumed days (`usedYearToDate`).
- **Brought Forward (Carryover) Card:** Unused annual leave carried from the previous year, showing active balance and dynamic Easter Sunday countdown (e.g. `Expires: 05 Apr 2026`).
- **Sick Leave Card:** Tracks days taken against the statutory 7.0-day allocation.
- **Compensation Leave (TOIL):** Displays approved non-expiring hours/days earned from weekend or holiday duty.
- **Monthly Accrual Progress:** Shows monthly accrual rate (1.75 days/mo) and next scheduled crediting date (1st of each month).
- **Quick Actions Hub:** 1-click access to Apply for Leave, Check Balances, Claim TOIL, Handover Coverage, and My Applications.

3. Leave Policies & Entitlement Framework

AgriLeave enforces statutory and company leave entitlements defined in `SYSTEM.LEAVE_POLICY`:

Leave Category (`SYSTEM.LEAVE_TYPES`)	Annual Entitlement	Accrual & Reset Rules	Carryover & Expiry Policy
Annual Leave	21 working days	Accrues at **1.75 days/month** on the 1st of each completed month.	Max 21 days can carry forward to the next year. Carried days expire on **Easter Sunday**.
Sick Leave	7 working days	Resets annually on **January 1st** to 7 days.	Non-cumulative (does not roll over). Consecutive leaves ≥ 2 days require a certified medical note. Can be applied retroactively.
Compensation (TOIL)	Earned as worked	Awarded when staff perform approved weekend or public holiday duties.	**Never expires**. Preserved indefinitely across calendar years.
Maternity Leave	90 calendar days	Granted per childbirth or adoption event.	Fully paid. Requires medical certificate confirming expected date of delivery.
Paternity Leave	14 calendar days	Granted per childbirth or adoption event.	Fully paid. Must be taken within 3 months of delivery.
Compassionate Leave	Up to 5 working days	Granted upon demise of immediate family member.	Covers spouse, child, parent, or sibling. Discretionary extension via HR.
Study Leave	Discretionary	Approved for recognized professional examinations and courses.	Requires supervisor sign-off and HR endorsement.

Leave Deduction Priority Order (checkLeaveBalanceWithPriority)

When an employee applies for Annual Leave, AgriLeave automatically consumes days in the following priority order: **1. Compensation days first** (awarded for weekend duty); **2. Brought-forward days second** (expiring pool — consumed before annual accruals to protect employee days); **3. Annual accrual pool third**; **4. Advance borrowing fourth** (against future monthly accruals if eligible).

4. Staff Guide: Applying for Leave

The leave application process is automated and intuitive. Business days are calculated dynamically as dates are selected, automatically excluding Saturdays, Sundays, and gazetted Kenyan holidays.

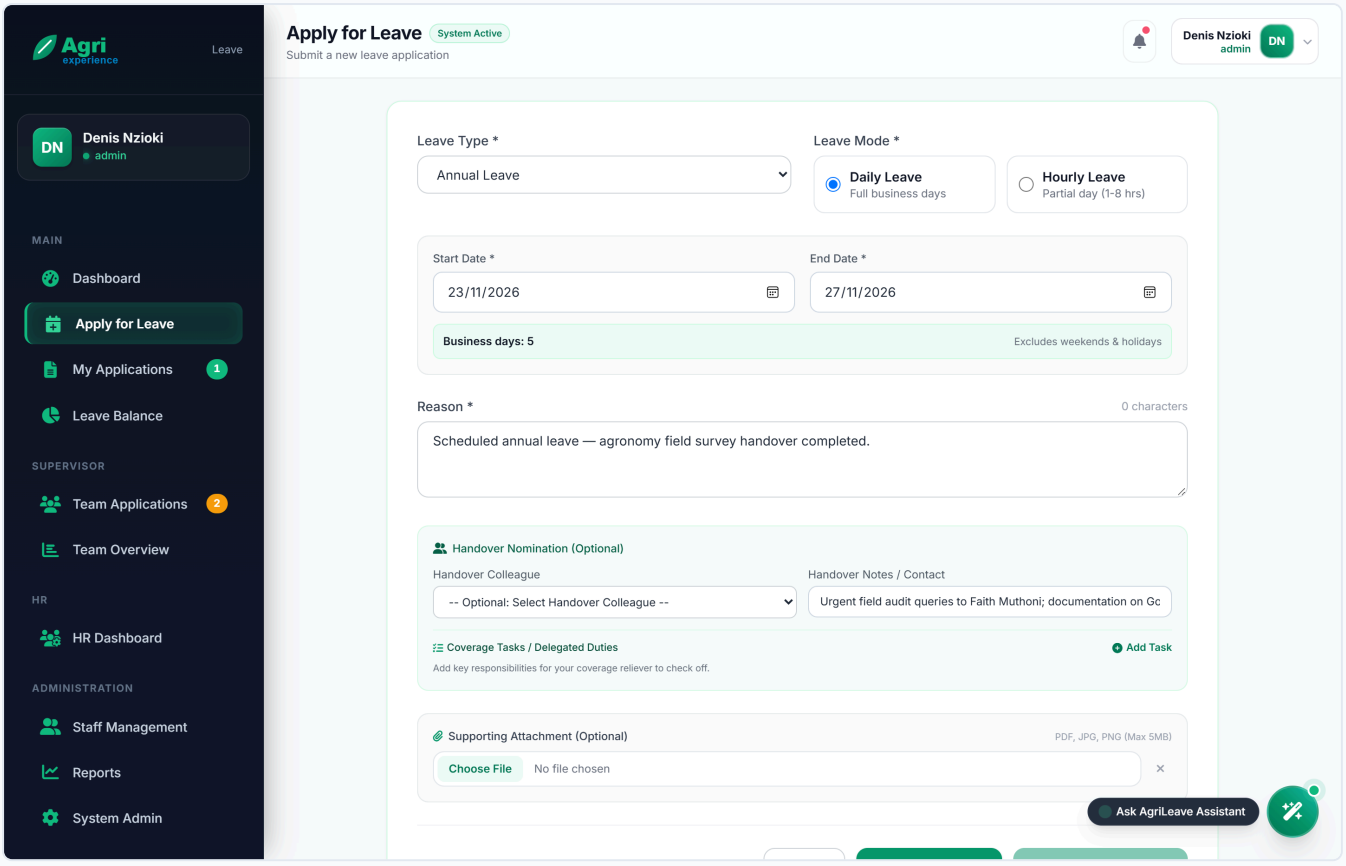


Figure 2: Leave Application Form with Daily/Hourly Modes, Duration Calculation, and Handover Nominee

Application Modes (SYSTEM.LEAVE_POLICY)

- **Daily Leave:** Standard full-day or multi-day requests. Minimum duration is **0.5 business days**. (Start date cannot be in the past, except for Sick Leave).
- **Hourly Leave:** Partial-day absences between **1 and 8 hours**. Converted to days at 8 working hours per day (e.g. 4 hours = 0.5 days; 2 hours = 0.25 days).

Step-by-Step Submission Instructions

1. **Select Leave Type & Mode:** Choose the applicable category. Toggle between *Daily Leave* (full days) or *Hourly Leave* (partial day 1–8 hours).
2. **Specify Dates:** Select Start Date and End Date. The system immediately displays the net deductible business days.
3. **Review Holiday Exclusions:** If your requested period spans a public holiday (e.g. Mashujaa Day, Jamhuri Day), an indicator confirms that the holiday is zero-rated and not deducted.
4. **Enter Reason:** Provide a clear, professional description of your absence (minimum 10 characters).
5. **Designate Handover Colleague & Tasks:** Select an operational peer who will provide coverage during your absence and add delegated duty checklist items.
6. **Upload Supporting Documents:** Required for Sick Leave (≥ 2 days) or specialized study/parental requests (PDF, JPG, PNG up to 5MB).
7. **Submit Application:** Click *Submit Leave Application*. Your supervisor and handover colleague receive immediate email alerts with 1-click response buttons.

5. Applications Tracking & Detailed Balances

AgriLeave provides complete transparency into all past and active applications, as well as an itemized mathematical breakdown of your accruals.

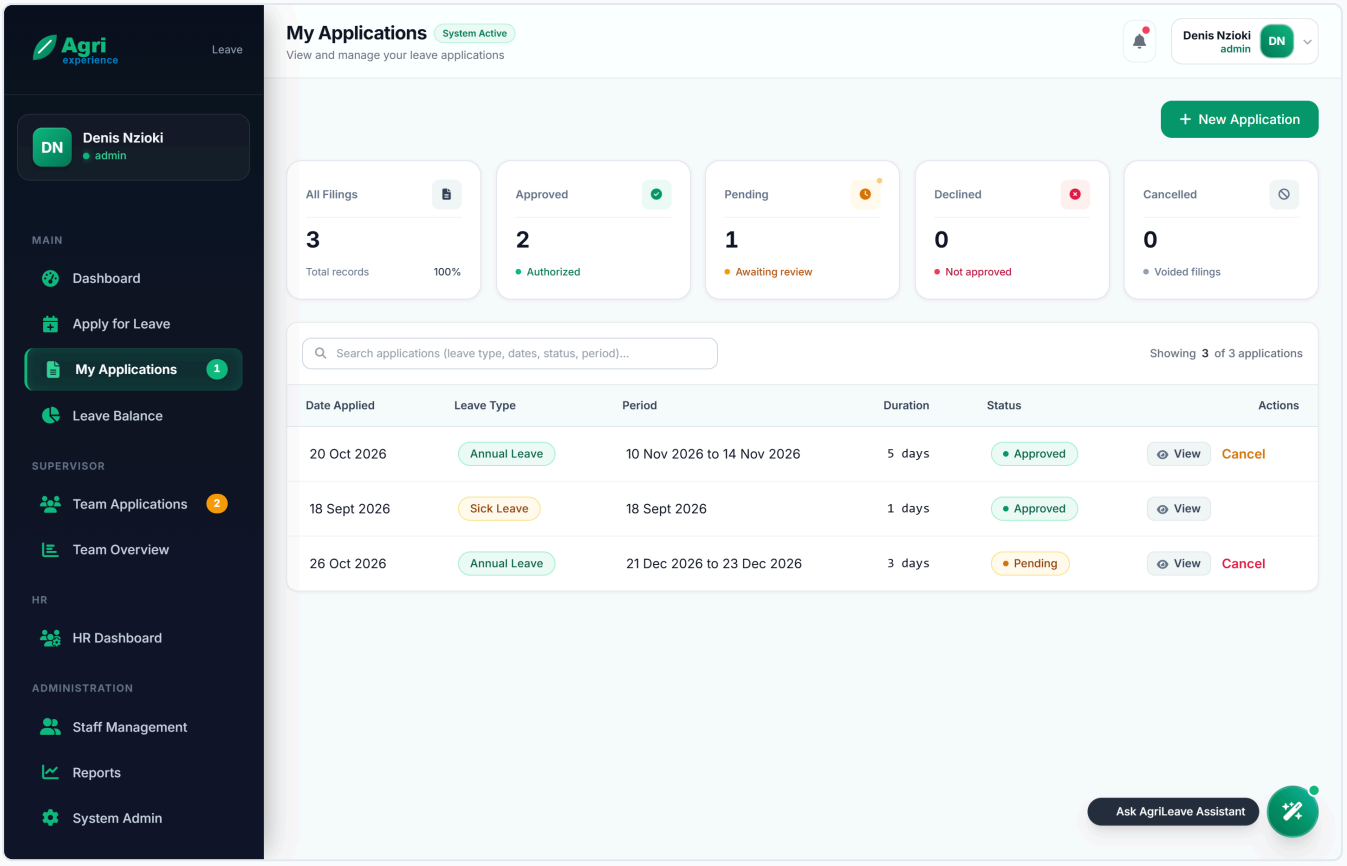


Figure 3: My Applications Table with Status Counters, Filters & Cancellation Controls

Tracking Applications

- **Status Counters:** Instant visibility into counts of *All*, *Approved*, *Pending*, *Declined*, and *Cancelled* applications.
- **Search Bar:** Instantly filter records by typing dates, leave types, or keywords.
- **Self-Service Cancellation:** Employees can cancel any request that has not yet been processed by clicking *Cancel* on the application row.

Detailed Leave Balance Breakdown

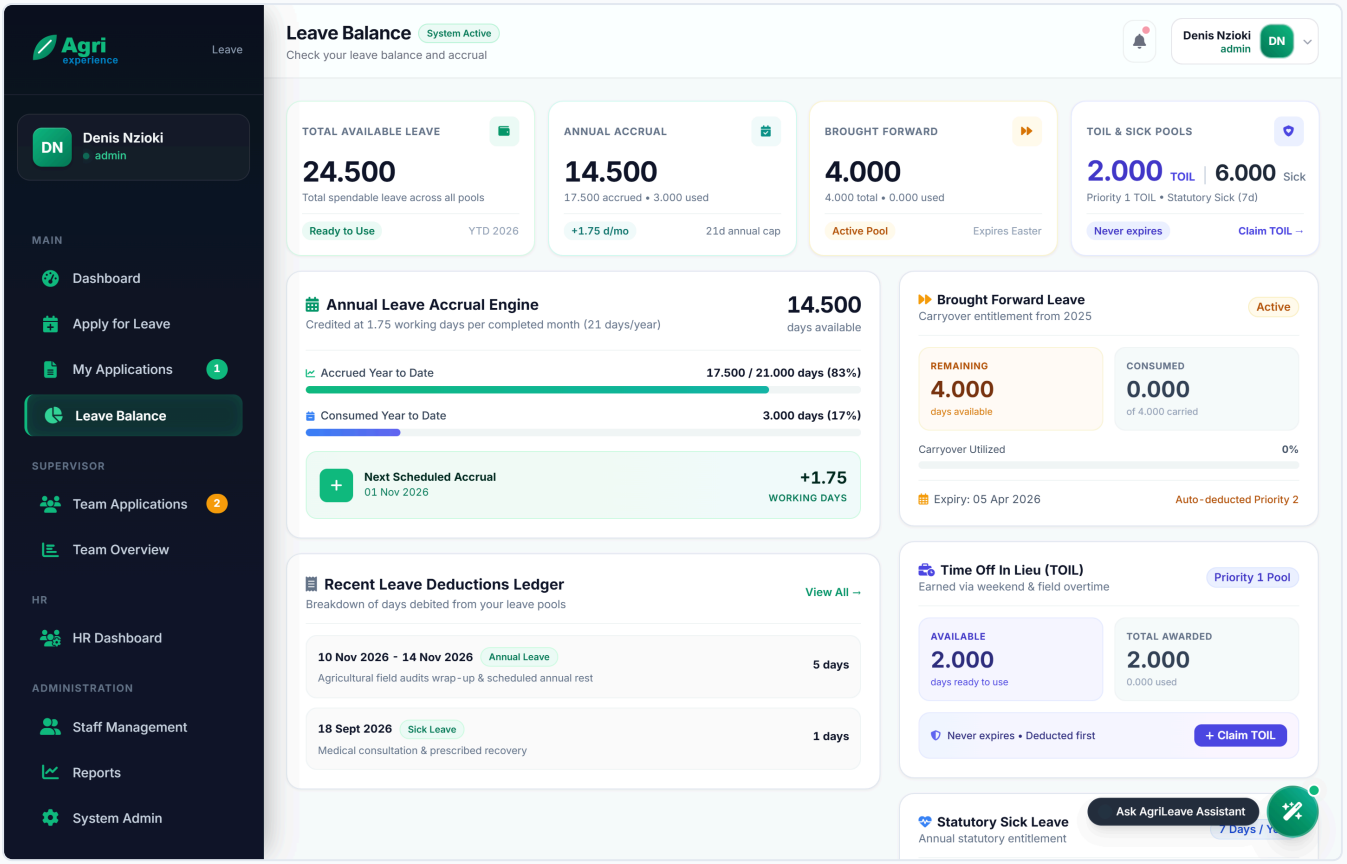


Figure 4: Detailed Balance Page with Monthly Accrual Tracker & Easter Expiry Status

The **Leave Balance** page provides full clarity on year-to-date accruals, unconsumed carryover balances with Easter countdowns, and compensation days earned.

6. Supervisor Guide: Review & Approvals

Supervisors possess specialized clearance to manage leave schedules for their direct reports, ensuring operational continuity across agricultural field projects.

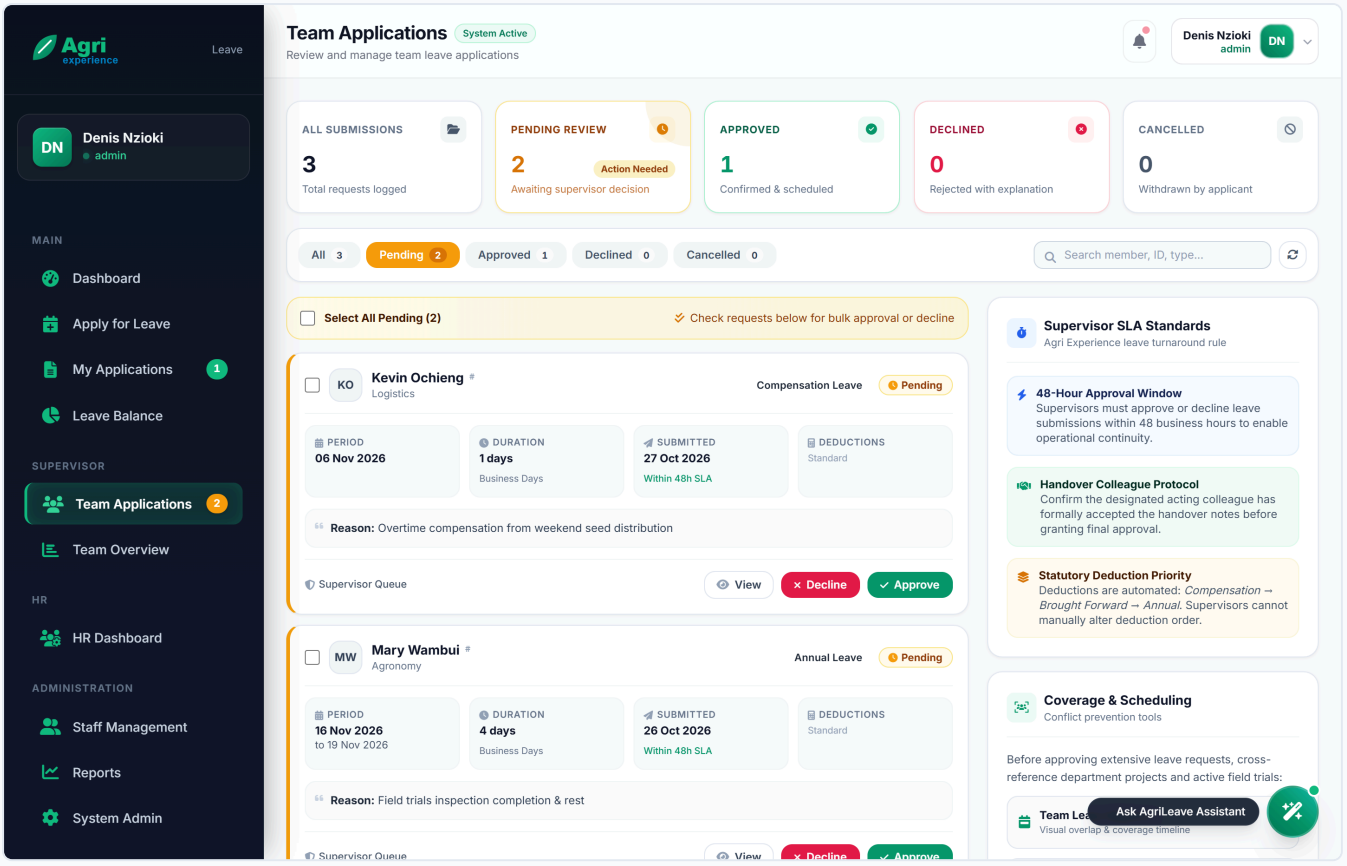


Figure 5: Supervisor Approval Center with Direct Report Queue, Clash Warnings & Batch Approvals

Approval Policies & Multi-Tier Routing

Leave Duration / Type	Approval Hierarchy	Routing Behavior (processApplicationDecision)
Standard Leave (≤ 5 days)	Single-Tier: Direct Supervisor	Supervisor approval immediately confirms leave, updates balances, sets Google Calendar event, and activates Gmail Vacation Responder.
Extended Leave (> 5 days)	Two-Tier: Supervisor + HR	Supervisor approves first (moves to `pending_hr`); request advances to HR queue for final operational clearance.
Specialized (Study, Parental)	Two-Tier: Supervisor + HR	Supervisor confirms departmental coverage; HR verifies compliance and certifications.

Supervisor Actions & Delegation

- **Clash Detection:** The queue automatically flags if another member of the same department is already scheduled off during the requested period.
- **Acting Supervisor Delegation:** If a primary supervisor delegates an acting supervisor via profile settings, the acting colleague can review and approve team applications during the delegation window.

- **1-Click Decisions:** Click *Approve* or *Decline*. Declinations prompt for a constructive reason included in the employee's notification.
- **Batch Approvals:** Use row checkboxes to approve seasonal queues simultaneously.

7. HR Dashboard & Staff Availability Wallchart

Human Resources administrators have company-wide clearance to oversee staffing density, monitor organization-wide liabilities, and execute payroll reconciliation.

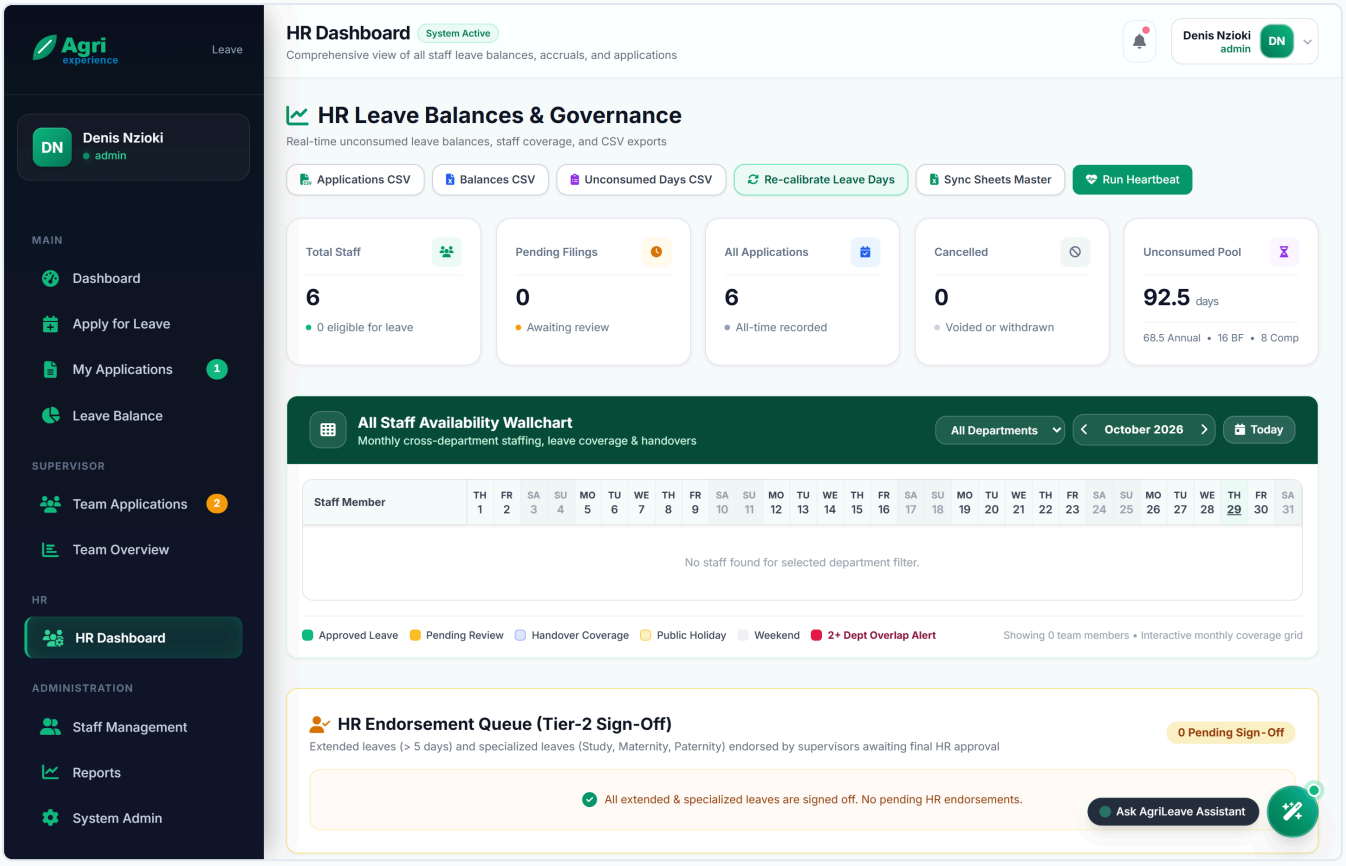


Figure 6: HR Governance Portal with Availability Wallchart, Liability Pool & CSV Exports

Key HR Governance Features

- **31-Day Availability Wallchart:** A comprehensive visual monthly wallchart displaying approved leaves, pending reviews, handover coverage duties, public holidays, and departmental overlap alerts.
- **Unconsumed Leave Pool (FinancialProvisionManager):** Live calculation of unconsumed leave liability days across Annual, Brought Forward, and Compensation pools for accounting provisioning.
- **HR Endorsement Queue:** Dedicated Tier-2 review stage for leaves exceeding 5 days and specialized leave requests.
- **One-Click CSV Exports:** Export Applications CSV, Balances CSV, or Unconsumed Days CSV for external reporting or payroll software integration.

8. System Administration & Maintenance

The **System Administration** interface provides system administrators with tools for annual lifecycle events, automated triggers, and policy controls.

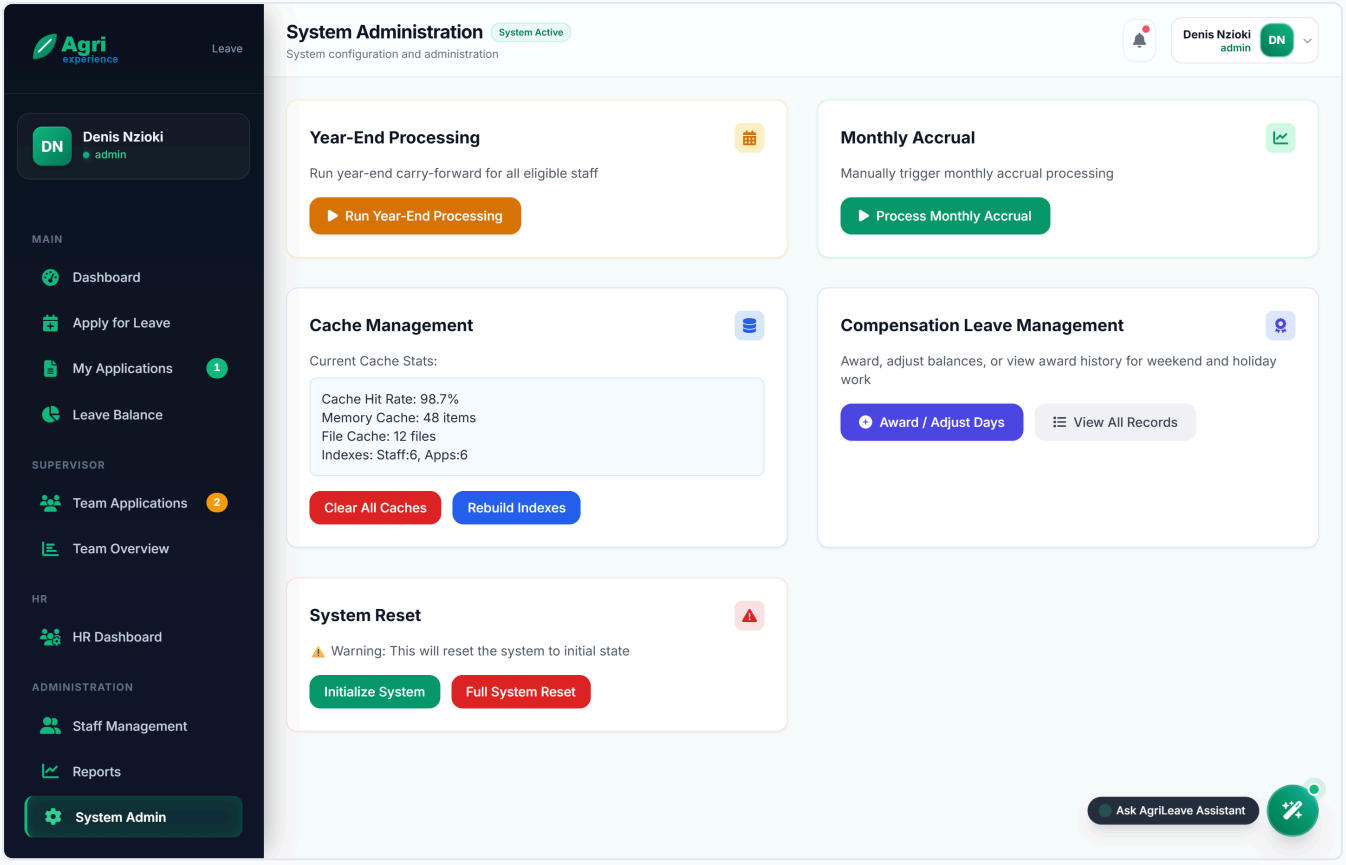


Figure 7: System Administration Console with Year-End Processing, Cache Analytics & Sync Controls

Core Administrative Tools

- **Year-End Processing Engine (AdminManager.processYearEnd):** Automated on January 1st at 00:30 (or run manually). Rollover unused annual leave up to 21 days into Brought Forward, stamps the new year's dynamic Easter expiration date, rolls over deficits, and resets sick allocations. Guarded by `yearEndProcessed_${year}`.
- **Monthly Accrual Trigger:** Credits 1.75 days to all eligible staff on the 1st of each month.
- **Operational Blackout Windows:** Configure restricted leave periods (e.g. KEPHIS seed certification audits or financial year-end) with strict or warning enforcement levels.
- **Sheets Database Mirror Sync:** One-click resynchronization between Google Drive JSON databases and the master Google Sheets workbook.
- **Kenya Holiday Recalibration:** When a new public holiday is gazetted, HR adds it and triggers recalibration to automatically refund working days to affected approved leaves.

9. Floating AI Assistant & Frequently Asked Questions

AgriLeave features a role-grounded floating AI Assistant widget accessible via the spark button in the bottom right corner of any page.

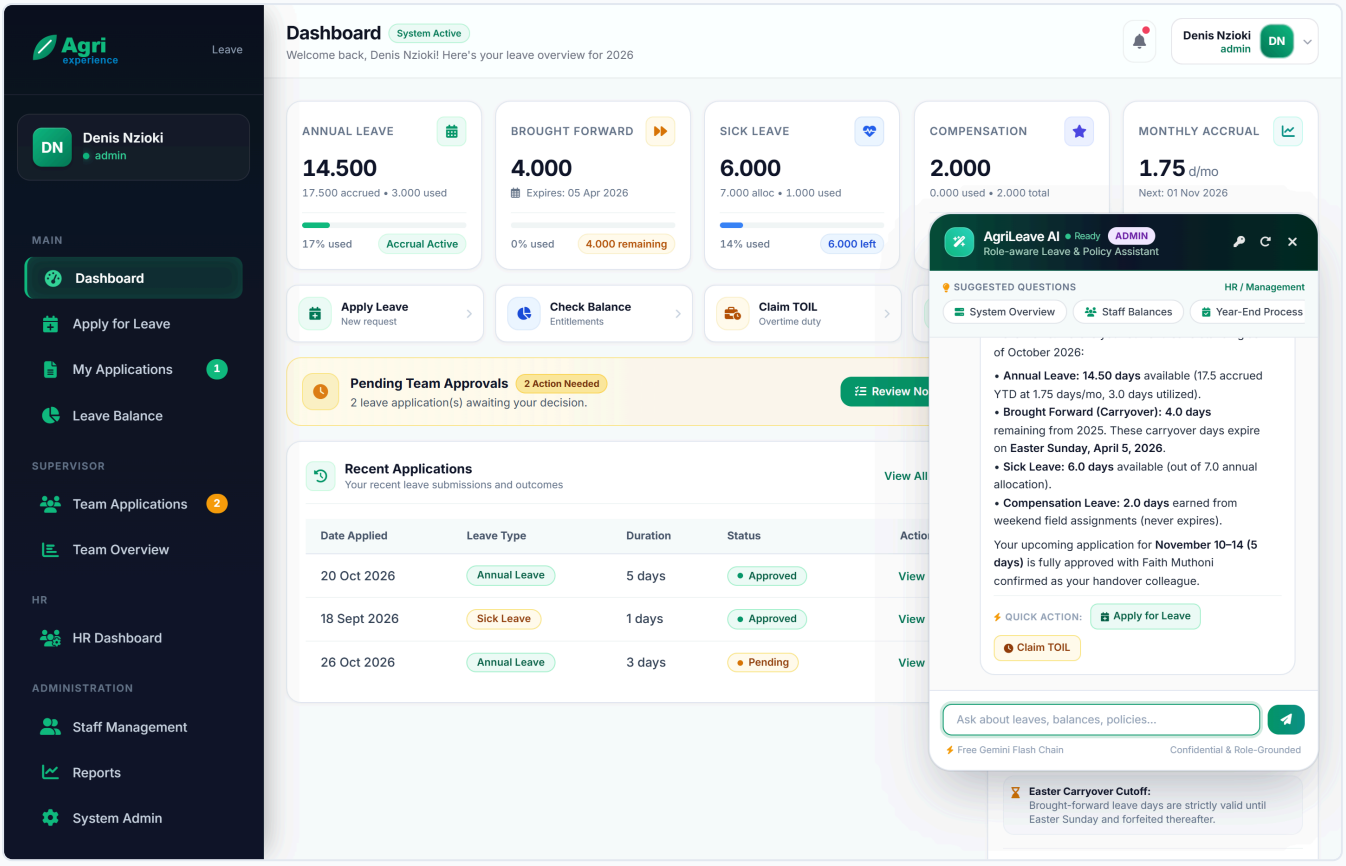


Figure 8: Floating Gemini AI Assistant with Role Grounding & Interactive Action Chips

AI Capabilities & Security Grounding (AIAssistantManager)

- **Strict Role Grounding:** Staff can only query their own balances and company policies. Supervisor and HR personas can query team metrics and organizational availability forecasts.
- **Multi-Model Fallback Chain:** Connects to Google AI Studio's free tier with automated failover: `gemini-3.5-flash-lite` → `gemini-3.5-flash` → `gemini-3.7-flash` → `gemini-3.8-flash`.
- **Interactive Action Chips:** Responses include 1-click action buttons (e.g. *Apply for Leave*, *Claim TOIL*) to transition directly into workflows.

10. Frequently Asked Questions (FAQ)

Q: How are weekends and Kenya public holidays treated?

A: AgriLeave automatically detects all weekends and Kenya gazetted holidays within your requested range and excludes them from leave deductions.

Q: What happens if I don't use my Brought Forward days by Easter?

A: Brought forward days carry an official expiration on Easter Sunday (calculated dynamically). Any unused balance is automatically forfeited after Easter.

Q: How do I earn Compensation Leave (TOIL)?

A: Staff performing approved duties on weekends or public holidays submit a Claim TOIL request. Once approved, hours convert to non-expiring leave days (8 hours = 1 day).

Q: Can Sick Leave be submitted retroactively?

A: Yes. While standard annual leave cannot have a past start date, Sick Leave allows retroactive submission upon recovery. Consecutive absences of 2 or more days require a medical note.